Incremental Multitask Contrastive Learning Network for End-to-End Few-Shot Open-Set Classification of Hyperspectral Images

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***Abstract*— Hyperspectral image (HSI) open-set classification (OSC) has gained increasing attention due to its practical significance. However, existing approaches face two major challenges: 1) poor and unstable classification performance under limited labeled samples and 2) the lack of end-to-end OSC frameworks. To address these issues, we propose an incremental multitask contrastive learning network (IMTCLN), which integrates four learning tasks to achieve end-to-end OSC under few-shot conditions through feature sharing and multitask collaboration. First, we introduce an expanded class labeling method in the model's output layer, enabling end-to-end OSC. Second, among the four learning tasks, the supervised classification task learns the mapping between known-class samples and their labels using limited labeled data. To enhance classification performance under few-shot conditions, we design a semi-supervised Euclidean contrastive learning task, which improves intra-class compactness and inter-class separability by modeling homogeneous and heterogeneous sample relationships. Additionally, for effective unknown-class recognition, we propose a supervised Mahalanobis contrastive learning task, optimizing the Mahalanobis distance among known classes to identify unknown-class samples. Finally, to further enhance classification stability, we introduce an incremental learning task, which leverages pseudo-labeled unknown-class samples to learn their discriminative features, enabling robust discrimination between known and unknown classes. Extensive experiments on three public datasets demonstrate that IMTCLN significantly outperforms existing methods, particularly under extremely limited labeled samples, showcasing superior OSC performance and stability.**

***Index Terms*— Hyperspectral image (HSI), multitask learning, open-set classification (OSC), semi-supervised learning.**

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I. Introduction

HYPERSPECTRAL imaging, as an advanced remote sensing technology, can capture hundreds of continuous and narrow spectral bands from visible light to infrared spectrum, and generate unique spectral reflection curves for different ground objects [1], [2]. With the development of technology, it can not only obtain rich spectral information but also have high spatial resolution. With these advantages, hyperspectral images (HSIs) have gradually emerged in many important fields, such as mineral exploration [3], agricultural management [4], water quality investigation [5], and so on. Among them, HSI classification (HSIC) is one of the most widely applied, which aims to assign pixels to corresponding classes [6], [7], [8], [9], [10], [11].

Over the past few years, a lot of HSIC methods [12], [13], [14], [15] have been proposed. The earliest classification methods are generally based on spectral classifiers, such as random forest (RF) [16], support vector machine (SVM) [17], and K-nearest neighbor (KNN). However, due to the problem of different objects with the same spectrum (DOSS), different spectra for the same object (DSSO), and nonlinearity in HSIs, these classical spectral classifiers that often ignore the spatial information are prone to obtaining a noisy classification map.

To overcome these problems, many HSIC methods based on deep learning have emerged and demonstrated significant advantages [18], [19], [20], [21]. They can not only extract spatial–spectral features through convolutional layers, but also extract deep abstract features that are beneficial for the classification layer by layer through supervised learning. For example, Roy et al. [22] proposed the spectral–spatial morphological attention transformer method, which integrates spectral and spatial features into the transformer framework to extract effective high-level features and improve classification accuracy. Lu et al. [23] proposed the SDGCP network, which uses attention mechanisms to effectively extract global features. Hu et al. [24] introduced a new multiscale multiangle attention network (MMAN) to model local and global features and reduce the complexity of self-attention. These methods provide excellent classification performance, but typically require a large amount of training data to train efficient models. In practice, the labels of HSIs are difficult to obtain due to the high collection cost [25], cumbersome labeling process, and significant time consumption [26], [27].

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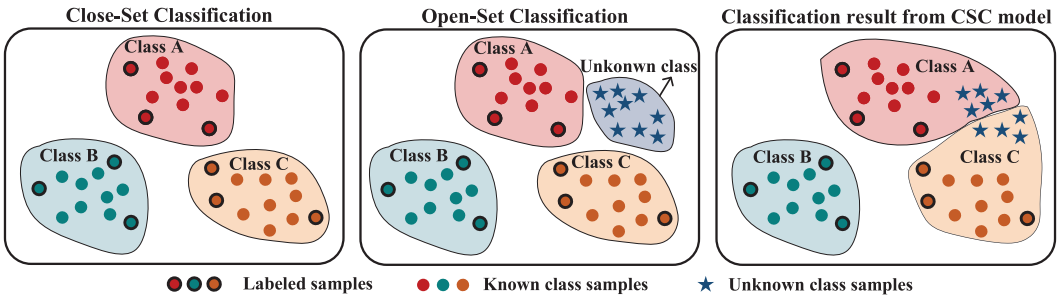


Fig. 1. Difference between close-set scene and OSC.

To address the challenge of label scarcity in HSIC, researchers have proposed various deep learning strategies with few labeled samples [28], [29], [30], [31], [32], [33], [34]. Wang et al. [35] proposed a new domain adaptive network with dual branches, which improved domain alignment and enhanced the model's ability to generalize from limited labeled data through the domain fusion and separation branches. Liu et al. [36] proposed a multitask learning framework, which effectively integrated auxiliary tasks such as cross-domain learning and contrastive learning into a network to extract more comprehensive semantic information. Braham et al. [37] introduced two positive pair mining strategies for contrastive learning, mitigating the need for high-quality data augmentations and providing an effective solution for few-shot HSIC. These methods have achieved superior performance in the traditional few-shot HSIC. However, these methods are usually based on the closed-set assumption requiring that training and testing data come from the same label space and cover all classes [38], which is often unrealistic.

Due to the fact that remote sensing images usually cover a wide geographical area, there are inevitably some uncommon, unexplored or unconsidered unknown classes, which are known as open-set environments [39], [40]. From the perspective of deep learning, the open-set problem means that there are new unknown class samples in the test set that are not encountered in the training set [41]. In this case, the traditional closed-set classification (CSC) methods prefer to classify the unknown classes into the known classes and cannot identify the unknown classes, as illustrated in Fig. 1. Therefore, an effective open-set classification (OSC) method is urgently needed, which should not only identify known classes but also detect unknown classes so that new classes can be rejected or marked as unknown classes when encountered [42], [43].

Currently, only a few works have been published on OSC of HSIs. Initially, some researchers used reconstruction loss or the distance between the prototype vector and the feature vector to fit a distribution, and then gave the probability that the sample was abnormal based on the extreme value theory. When the probability exceeds a certain threshold, the sample is considered an unknown class [41], [44], [45]. These methods have initially achieved OSC of HSIs, which have pioneering value. In essence, these methods are based on the assumption that the training set and test set of known classes have the same distribution. However, due to the existence of DOSS and DSSO in HSIs, this assumption is often not always true in actual complex scenes.

Later, in order to avoid these problems, some other OSC methods with different concepts were proposed for HSIs [46], [47], [48]. For instance, Du et al. [43] proposed a weighted Pearson distance metric designed for the identification of unknown samples in open-set scenarios. Sun et al. [49] proposed the reciprocal points learning to enhance performance in open-set environments by modeling extra-class space and utilizing a learnable dynamic threshold. Ji et al. [50] first introduced evidential learning into the OSC tasks, embedding domain adaptation into the generative adversarial network to generate unknown class samples. The common problem of these methods is that they all rely on thresholds to identify unknown classes, but threshold-based methods usually lack flexibility, robustness, and accuracy. For this purpose, Pal et al. [51], [52] proposed a novel outlier calibration network (OCN) to avoid the need for manual thresholding. These methods have achieved good performance in OSC of HSIs, but there are still some limitations.

1) They still need a small number of labeled samples to achieve superior performance, and have not yet solved the OSC problem under extremely limited labeled samples (only a few labels per class).
2) They usually discriminate unknown classes by learning known classes and lack of learning of unknown classes, so that the recognition of unknown classes is limited.
3) Most of the existing OSC methods for HSIs require further processing of the network output results, but rarely focus on the end-to-end open-set recognition problem.

To overcome the above problems, we propose a novel incremental multitask contrastive learning network (IMTCLN) for end-to-end few-shot OSC of HSIs. The core advantages of IMTCLN include are follows.

1) It achieves robust few-shot OSC through synergistic multitask collaboration.
2) It actively learns unknown class features through an incremental learning mechanism, significantly improving the accuracy and stability of unknown class recognition.
3) It realizes end-to-end open-set classification through extended class labeling (ECL).

Specifically, first, a network structure consisting of two parameter-sharing feature extractors and three

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parameter-sharing classifiers is designed, which extracts features from two different training sets. Then, the network is trained under the constraints of three learning tasks designed in this article: supervised classification, semi-supervised Euclidean contrastive learning, and supervised Mahalanobis classification learning. Subsequently, when the network is trained to a relatively stable state under the above three learning tasks, an incremental learning task is designed to constrain the network training until the network re-entered a stable state under the four tasks. Finally, all samples that need to be predicted are sent to the trained network, and the predicted label results belonging to known and unknown classes are directly output.

Mahalanobis distance has the advantages of measuring feature correlation and scale difference, while Euclidean distance is simple to calculate, geometrically intuitive, and robust to estimation errors. This article combines the two metric spaces of Euclidean and Mahalanobis distance to constrain the network to learn not only statistical sample relationships but also subtle sample differences. This joint constraint with good computational efficiency and robustness can effectively improve the performance of classification.

The major contributions of this study are concluded as follows.

1) An IMTCLN is proposed. Through feature sharing and synergistic multitask learning, it can not only overcome poor OSC performance under extreme sample scarcity, but also realize end-to-end OSC.
2) To enhance the classification performance under limited labels, a semi-supervised Euclidean contrastive learning task is designed to constrain the predicted probability of the label space, which aims to improve the ability to discriminate intra-class and inter-class samples by learning the homogeneous and heterogeneous relations between samples.
3) In order to effectively identify unknown classes, a supervised Mahalanobis contrastive learning task is designed, which optimizes the Mahalanobis relationship between samples of known classes in the feature space, thereby detecting unknown classes through a unknown class identification strategy (UCR) based on Mahalanobis distance.
4) To further improve the accuracy and stability of unknown class recognition, an incremental learning strategy is proposed, which is initiated when the old task converges. This strategy uses the discovered pseudo-unknown classes from the old task to learn the features of the unknown classes.

The remaining structure of this article is as follows. Section II describes the proposed method. Section III shows the dataset setup, experimental results, and experimental analysis. Section IV provides the conclusion.

II. Proposed Method

In this article, we denote an HSIs open-set data as $\mathbf{X} \in \Re^{n\times D} = \{x_1, \ldots, x_n\} = \{\mathbf{X}^L, \mathbf{X}^U = \{\mathbf{X}^{U_k}, \mathbf{X}^{U_u}\}\}$, where $\mathbf{X}^L = \{x_1^L\}, \ldots, x_{n_L}^L\}$ represents the labeled sample set with labels $\mathbf{Y} = \{y_1, \ldots, y_{n_L}\}$ and $\mathbf{X}^U$ represents the unlabeled sample set. Additionally, $\mathbf{X}^{U_k}$ and $\mathbf{X}^{U_u}$ denote the unlabeled known and unknown class samples within the open-set data, respectively. The target task of the proposed IMTCLN is to effectively detect unknown class samples $\mathbf{X}^{U_u}$ from unlabeled samples $\mathbf{X}^U$ and correctly classify known class samples $\mathbf{X}^{U_k}$ based on the extremely limited labeled samples $\mathbf{X}^L$.

A. Overall Framework

The overall framework of the proposed IMTCLN for end-to-end few-shot OSC of HSIs is shown in Fig. 2. The network structure includes two parameter-sharing feature extractors (I, II) and three parameter-sharing classifiers (I, II, III), designed to address four learning tasks: supervised classification, semi-supervised Euclidean contrastive learning, supervised Mahalanobis classification learning, and incremental learning.

First, to mitigate the overfitting issue caused by limited labeled samples, the two feature extractors are trained under two distinct training sets: training set A (containing only labeled samples) for two supervised learning tasks, and training set B (containing both labeled and unlabeled samples) for semi-supervised and incremental learning tasks. Subsequently, the output features of the two feature extractors are mapped to the label space to generate class probability values after a series of operations of three classifiers (such as dropout, linear, and softmax activation functions). Then, the output features from the feature extractor I and the output probabilities from the three classifiers are each constrained by the four learning tasks.

Among them, the supervised classification task guides classifier I to learn class label information under the supervision of labeled samples. The semi-supervised Euclidean contrastive learning task constrains classifier II to learn the spectral-locational-spatial Euclidean relationship between all samples (including labeled and unlabeled samples) to alleviate the overfitting problem and poor classification performance under extremely limited labeled samples. The supervised Mahalanobis classification task is to optimize the Mahalanobis relationship between labeled samples of known classes based on the output features of feature extraction I. When the above three tasks are optimized to a relatively stable state, the incremental learning task is started to constrain the output probability of classifier III to learn the unknown class pseudo-label derived based on the first three tasks. The proposed IMTCLN is trained under the synergistic constraints of the above four complementary learning tasks. Finally, all unlabeled samples are input into the trained IMTCLN to output the probabilities of belonging to multiple known classes and unknown classes. The details of IMTCLN will be introduced later.

B. Network Structure

The network structure of IMTCLN consists of two feature extractors with the same structure and shared parameters, and three classifiers with the same structure and shared parameters.

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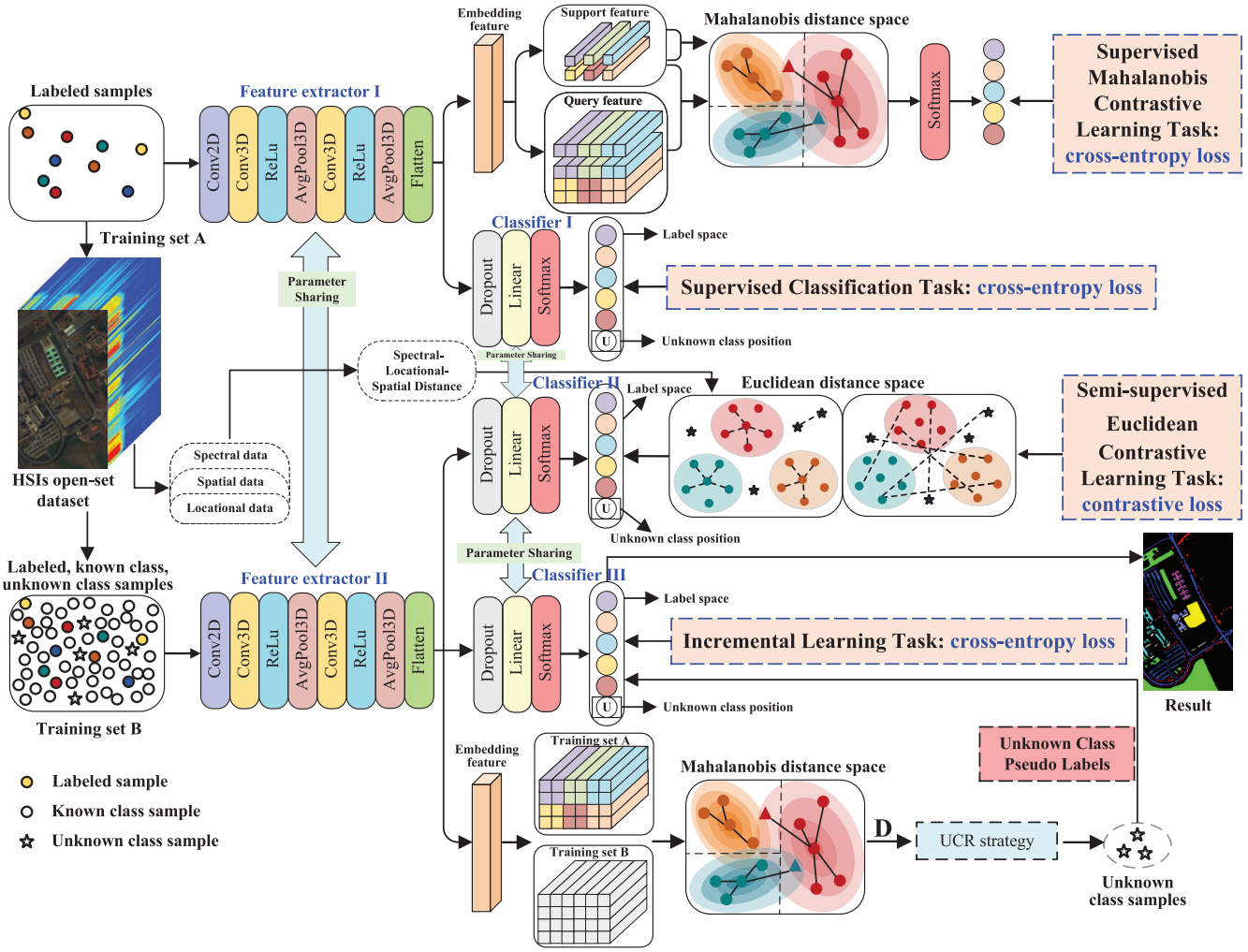


Fig. 2. Framework of the proposed IMTCLN for end-to-end few-shot OSC of HSIs.

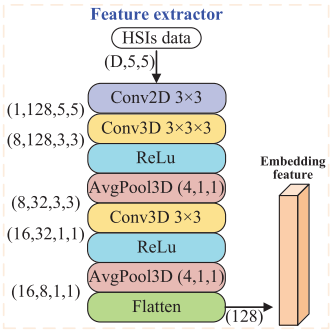


Fig. 3. Framework of the feature extractor.

Spatial information has been proven to be beneficial for HSIC. To effectively extract spatial-spectral features, the feature extractor consisting of 2-D and 3-D convolutional layers is designed, as shown in Fig. 3. First, smaller 3×3 convolutional kernels are used to capture fine spatial variations. Second, 3-D convolution is employed to capture contextual differences in the spectral dimension. By sequentially linking 2-D and 3-D convolutions, the network effectively combines spatial and spectral features. In addition, two average pooling layers are placed to reduce the number of parameters of the model to mitigate the risk of overfitting. The ReLU activation function is used to increase the nonlinear representation capabilities, allowing for the learning of more complex patterns [41]. Finally, the output of the average pooling layer is flattened for subsequent processing of the embedded features extracted by the feature extractor.

The two feature extractors of IMTCLN extract features from different datasets, and their output features can be expressed as

$$\begin{aligned} \mathbf{F}_{\mathrm{I}} &= \left\{f_1^L, \ldots, f_{n_L}^L\right\} = f_{\mathrm{I}}\left(\mathbf{X}^L, \Theta_F\right) \\ \mathbf{F}_{\mathrm{II}} &= \{f_1, \ldots, f_n\} = f_{\mathrm{II}}(\mathbf{X}, \Theta_F) \end{aligned} \tag{1}$$

where the functions f_{I} and f_{II} represent the mapping performed by feature extractors I and II, respectively. $\mathbf{X}^L$ is the training set A containing only labeled samples, and $\mathbf{X}$ is the training set B containing all labeled and unlabeled samples. Θ_F is the set of network parameters shared by both feature extractors.

Next, the output features of the two feature extractors are fed to three classifiers with the same structure. The three classifiers map the features $\mathbf{F}_{\mathrm{I}}$ and $\mathbf{F}_{\mathrm{II}}$ to the label space

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based on *Dropout* and *Linear*, where Classifiers I and III output normalized prediction probabilities through the *Softmax* function. It should be emphasized that in order to achieve end-to-end recognition of unknown classes, we adopted the ECL method, which adds a node in the label space of the network output as the position to represent the probability of belonging to the unknown class. This additional node is used to generate the pseudo-label for the unknown class, which is then incorporated into the training process to enhance the ability to recognize and adapt to new unknown classes. The output probabilities of the three classifiers can be expressed as

$$\begin{aligned}\mathbf{P}_{\mathrm{I}} &= \left\{p_1^L, \ldots, p_{n_L}^L\right\} = c_{\mathrm{I}}(\mathbf{F}_{\mathrm{I}}, \Theta_C)\\ \mathbf{P}_{\mathrm{II}} &= \{p_1, \ldots, p_n\} = c_{\mathrm{II}}(\mathbf{F}_{\mathrm{II}}, \Theta_C)\end{aligned} \tag{2}$$

where c_{I} and c_{II} represent the mapping operations of the classifier I and II, respectively. Θ_C represents the learning parameters shared by the three classifiers. Since classifier III, corresponding to the incremental learning task, needs to wait until the other three tasks have learned to a stable stage before it can represent and learn data, we will not explain it here.

Finally, the features and probabilities output above are constrained for different learning tasks.

C. Supervised Classification Task

Like the tasks of traditional classification networks, the supervised classification task is based on labeled samples to guide the network to learn the mapping between samples and class labels to achieve classification. In fact, it is the basis for achieving end-to-end classification. It is to find the optimal parameters for IMTCLN by minimizing the cross-entropy loss function L_{sc}

$$L_{\mathrm{sc}}(\mathbf{Y}, \mathbf{P}_{\mathrm{I}}) = -\frac{1}{n_L}\sum_{i=1}^{n_L}\sum_{c=1}^{C+1} y_{ic}\log\left(p_{ic}^L\right) \tag{3}$$

where y_{ic} is the label of the ith labeled sample on the cth class. p_{ic}^L is the probability that the ith labeled sample belongs to the cth class. C is the number of known classes, and $C+1$ represents the number of classes after adding unknown classes.

D. Semi-Supervised Euclidean Contrastive Learning Task

For HSIs, labeled data is usually very expensive and difficult to obtain, but unlabeled data is relatively easy to obtain and has a large quantity. In order to solve the problem of overfitting and poor classification performance of the network under extremely limited labeled samples, a semi-supervised Euclidean contrastive learning task based on all samples (including labeled and unlabeled samples) is designed, which guides the network to learn the Euclidean relationship between samples in a semi-supervised manner.

In fact, this task is designed based on the graph and expresses the relationship between the vertices of the graph. First, we need to construct an undirected graph $\mathbf{G} = \{\mathbf{X}, \mathbf{E}, \mathbf{W}\}$ of HSIs, where $\mathbf{X} = \{x_1, \ldots, x_n\}$ represents the vertex set and corresponds to n samples of the HSIs dataset. $\mathbf{E} = \{(x_i, x_j)|x_i \in \mathbf{X}, x_j \in \mathbf{X}\}$ represents the edge set and indicates whether there is a connection between samples of HSIs. $\mathbf{W} = \{W_{ij}|x_i \in \mathbf{X}, x_j \in \mathbf{X}\}$ represents the edge weight matrix and is also the weight set of the relationship between samples. The key challenge in building this graph is to establish the edges connecting the vertices. This problem has been solved in our previous research [53], which proposed an effective method for measuring the similarity (Spectral-Locational-Spatial Distance) between HSI samples to achieve edge selection and weight calculation.

Generally, the nearest neighbor relationship of the graph is used to participate in the processing of HSIs. However, under the training condition of extremely limited labeled samples, using only the nearest neighbor relationship within the graph as a network constraint will cause the representation distance between all samples to become smaller and smaller, which will reduce the discriminability of sample representation and increase the difficulty of classification. To tackle this issue, we propose to construct two graphs: a homogeneous graph $\mathbf{G}^n = \{\mathbf{X}, \mathbf{E}^n, \mathbf{W}^n\}$ that expresses the nearest neighbor relationship and a heterogeneous graph $\mathbf{G}^f = \{\mathbf{X}, \mathbf{E}^f, \mathbf{W}^f\}$ that expresses the farthest neighbor relationship.

To enhance the quality of the two graph representations and effectively learn potential structural information of HSIs, we make full use of the extremely limited labels in HSI dataset to optimize the edges. In the homogeneous graph, the labeled samples of the same class are preferentially connected and the maximum weight is set, while the labeled heterogeneous or relatively low similarity neighbors in the neighborhood are discarded. The opposite is true for the heterogeneous graph. So these two graphs are actually semi-supervised.

In practical applications, we use the nearest neighbors and farthest neighbors in the two graphs to guide network optimization. In order to make the relationship in the two graphs more effective in promoting classification and propagating labels in unlabeled samples, we directly apply it to the output of the classifier II, and its contrastive loss function can be expressed as

$$\begin{aligned}L_{\mathrm{ssecl}} = &\frac{1}{n}\sum_{i=1}^{n}\sum_{x_j\in N^n(x_i)} W_{ij}^n\left\|p_i - p_j\right\|^2\\ &+\exp\left(-\frac{1}{n}\sum_{i=1}^{n}\sum_{x_j\in N^f(x_i)} W_{ij}^f\left\|p_i - p_j\right\|^2\right)\end{aligned} \tag{4}$$

where $N^n(x_i)$ and $N^f(x_i)$ are the sample sets connected to sample x_i in $\mathbf{E}^n$ and $\mathbf{E}^f$, respectively, representing the nearest and farthest neighbors of x_i. W_{ij}^n and W_{ij}^f are their connection weights. p_i and p_j are the output probabilities of sample x_i and sample x_j on classifier II, respectively.

L_{ssecl} encourages homogeneous samples to have similar prediction probabilities while expanding the distance between the prediction probabilities of heterogeneous samples. This not only accelerates the process of unlabeled known-class samples finding their correct class labels, thereby improving the classification accuracy of known classes, but also enhances the recognition of unknown classes by separating the prediction probabilities of unknown classes from known classes.

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E. Supervised Mahalanobis Classification Task

Current OSC methods for HSIs [43], [52] usually rely on the distance estimation between sample features and class prototypes to detect outliers as unknown class samples. However, this metric has two main limitations: 1) it ignores the correlation and amplitude variation between dimensions and 2) it ignores different intra-class distributions and inter-class decision boundaries. To address this, Pal et al. [54] proposed Prototype OpenMax (P-OpenMax) by introducing distance calibration to address the shape and scale differences between feature vectors, but it ignores the decision boundary differences between classes.

To address the above limitations, we propose to use the Mahalanobis distance to estimate the distance between sample features. It measures the correlation and intensity difference between feature components through the covariance matrix, which can effectively eliminate the correlation and scale uniformity by projecting the feature space into the Mahalanobis space. In addition, it can also promote the intra-class compactness and inter-class separability of prototype features through supervised Mahalanobis distance training, further improving the accuracy of unknown class recognition. The specific formula is as follows:

$$d_{c,i}\left(f_c^{\mathbf{S}^L}, f_i^{\mathbf{Q}^L}\right) = \left(f_c^{\mathbf{S}^L} - f_i^{\mathbf{Q}^L}\right)^T (\mathbf{U}_c)^{-1}\left(f_c^{\mathbf{S}^L} - f_i^{\mathbf{Q}^L}\right) \quad (5)$$

where $\mathbf{U_c}$ is the covariance matrix of the cth class, estimated from the feature vectors of the support set. The support set $\mathbf{S}^L = \{x_1^L, \ldots, x_c^L, \ldots, x_C{}^L\}$ is a subset of $\mathbf{X}^L$, containing one sample per class. The query set $\mathbf{Q}^L$ is the complement of the support set S within the labeled sample set $\mathbf{X}^L$, that is, $\mathbf{Q}^L = \{x \in X^L | x \notin S^L\}$.

In order to eliminate both intra-class and inter-class correlation and scale differences, the covariance matrix $\mathbf{U}_c$ is a weighted combination of the intra-class covariance matrix Σ_c, the inter-class covariance matrix Σ, and the identity matrix $\mathbf{I}$, as shown below [38]

$$\mathbf{U}_c = \frac{1}{2}\Sigma_c + \frac{1}{2}\Sigma + \mathbf{I}. \quad (6)$$

In order to enhance the recognition of unknown classes, we minimize intra-class prototype distances while maximizing inter-class separation. Concretely, we use the supervised Mahalanobis classification loss L_{smc} to optimize the support set feature vectors (i.e., prototype vectors) distribution as follows:

$$p_i^L\left(y = c \middle| x_c \in S^L, x_i \in Q^L\right) = \frac{\exp\left(-d_{c,i}\left(f_c^{S^L}, f_i^{Q^L}\right)\right)}{\sum_{k=1}^{C}\exp\left(-d_{k,i}\left(f_k^{S^L}, f_i^{Q^L}\right)\right)} \quad (7)$$

$$\mathbf{L}_{\text{smc}} = -\frac{1}{\left|\mathbf{Q}^L\right|}\sum_{i=1}^{|\mathbf{Q}^L|}\sum_{c=1}^{C} y_{ic}\log\left(P_i^L(y=c)\right) \quad (8)$$

where p_i^L is the probability that the ith labeled sample belongs to the cth class. The subscript L indicates that all samples in (7) and (8) are drawn from labeled data.

In order to find potential unknown class samples in the training set B, we calculate the Mahalanobis distance matrix $\mathbf{D} = (\mathbf{S}, \mathbf{Q})$ with the training set A as the support set $\mathbf{S}$ to calculate the prototype vector and B as the query set $\mathbf{Q}$. The specific formula is as follows:

$$\mathbf{D} = \left[d_{c,i}\left(f_c^{\mathbf{S}}, f_i^{\mathbf{Q}}\right)\right]_{\substack{c=1,\ldots,C \\ i=1,\ldots,n}} = \begin{bmatrix} d_{1,1}\left(f_1^{\mathbf{S}}, f_1^{\mathbf{Q}}\right) & \cdots & d_{C,1}\left(f_C^{\mathbf{S}}, f_1^{\mathbf{Q}}\right) \\ \vdots & \ddots & \vdots \\ d_{1,n}\left(f_1^{\mathbf{S}}, f_n^{\mathbf{Q}}\right) & \cdots & d_{C,n}\left(f_C^{\mathbf{S}}, f_n^{\mathbf{Q}}\right) \end{bmatrix} \quad (9)$$

where $d_{c,i}(f_c^{\mathbf{S}}, f_i^{\mathbf{Q}})$ denotes the Mahalanobis distance between the feature vector of the ith sample and the prototype vector of the cth class.

Motivated by these observations, we design a UCR to effectively identify potential unknown class samples by analyzing the Mahalanobis distance matrix $\mathbf{D}$. Since the UCR strategy is used in incremental learning, we will provide a detailed introduction in the Section II-F.

F. Incremental Learning Task

Existing OSC methods for HSI usually rely on the boundary information of known classes and cannot model the feature distribution of unknown classes, resulting in poor recognition accuracy and stability. To solve this problem, we propose an incremental learning task to expand the classification ability of the model by gradually learning the features of unknown classes. Specifically, the incremental learning task is to learn the mapping between potential unknown class samples and their pseudo-labels from the UCR strategy by constraining the output probability of classifier III. IMTCLN therefore has the capability of end-to-end OSC. The detailed process is as follows:

First, the output probabilities of the classifier III can be expressed as

$$\mathbf{P}_{\text{III}} = \left\{p'_1, \ldots, p'_n\right\} = c_{\text{III}}(\mathbf{F}_{\text{II}}, \Theta_C). \quad (10)$$

Second, as shown in Fig. 4, during the training process, the UCR strategy is actually used for each batch of data, so $\mathbf{D} = [d_{c,i}(f_c^{\mathbf{S}}, f_i^{\mathbf{Q}})]_{\substack{c=1,\ldots,C \\ i=1,\ldots,Bz}}$. The UCR strategy aims to select samples with a high probability of not belonging to a known class as potential unknown classes by analyzing the Mahalanobis distance between the data in each batch of data and each class prototype, and assigning pseudo-labels to the unknown classes. The UCR strategy is as follows:

$$\delta = \{\delta_i\}_{i=1}^{Bz} = \left\{\min_c \mathbf{D}(c,i)\right\}_{i=1}^{Bz} = \left\{\min_{1\le c\le C} d_{c,i}\left(f_c^{\mathbf{S}}, f_i^{\mathbf{Q}}\right)\right\}_{i=1}^{Bz} \quad (11)$$

where Bz represents the batch size of the training set B and δ_i means the distance between the ith sample feature and the nearest class prototype in a batch of data. Typically, larger δ_i indicates a higher likelihood of x_i being from the unknown class.

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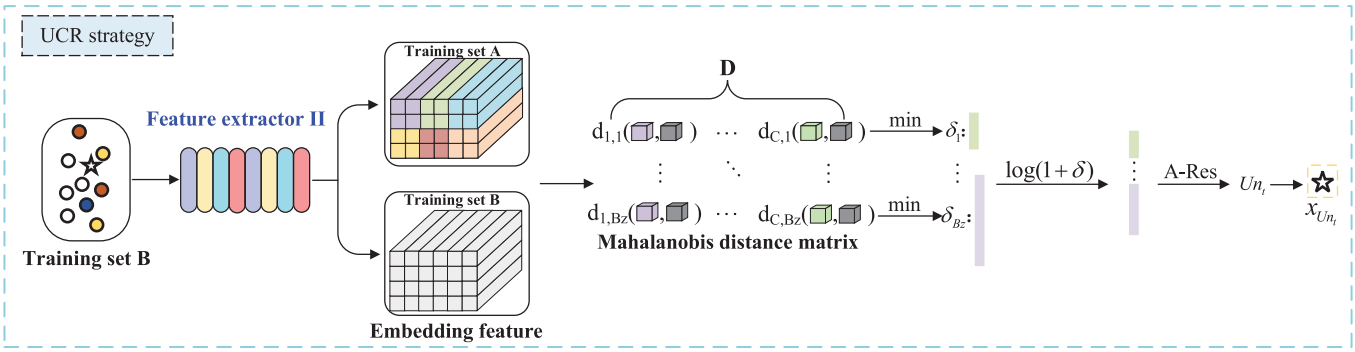


Fig. 4. Framework of the UCR.

In fact, through many experiments, we found that directly selecting the maximum value sample in δ as the unknown class sample is not conducive to the detection of multiple unknown classes. For this reason, we choose to use the distance value in δ as the weight and perform weighted random sampling through the A-Res algorithm (the sampling probability is proportional to the weight) [55]. To avoid excessive sampling bias toward extreme outliers, we first apply a log transformation to smooth the distance values using $\log(1+\delta)$. Then, A-Res algorithm is used to output the sampling index Un_t as the unknown class sample index for the current training. The formula is as follows:

$$Un_t = \text{A-Res}(\log(1+\delta)). \tag{12}$$

The UCR strategy can dynamically find potential unknown class samples x_{Un_t} in each training batch. At the same time, the network is trained and optimized based on these unknown class samples. It is important to note that these unknown class samples are only effective for the network optimization of the current epoch, and new unknown class samples will be searched again in each epoch.

Finally, the pseudo-label of the unknown class can guide the classifier III to extract the features of the unknown class from x_{Un_t}

$$L_{\mathrm{il}}^t = \begin{cases} -\sum_{c=1}^{C+1} y_c^{\mathrm{Un}} \log\left(p'_{Un_t,c}\right), & L_{\mathrm{il}}^t \le \dfrac{\gamma}{t-T}\sum_{k=T}^{t-1} L_{\mathrm{il}}^k \\ -\alpha \sum_{c=1}^{C+1} y_c^{\mathrm{Un}} \log\left(p'_{Un_t,c}\right), & L_{\mathrm{il}}^t > \dfrac{\gamma}{t-T}\sum_{k=T}^{t-1} L_{\mathrm{il}}^k \end{cases} \tag{13}$$

where p'_{Un_t} means the predicted probability output of the unknown class sample x_{Un_t}, and its corresponding pseudo-label of the unknown class is y^{Un}. It is worth noting that when the loss L_{il}^t exceeds the threshold of the historical average loss $(\gamma/t-T)\sum$, the loss is suppressed by the weight α, thereby stabilizing the pseudo-label learning process of the unknown class sample.

In summary, the total loss of IMTCLN in this article can be expressed as

$$\mathcal{L}(t) = \begin{cases} L_{\mathrm{sc}} + L_{\mathrm{ssecl}} + L_{\mathrm{smc}}, & \text{if } t \le T \\ L_{\mathrm{sc}} + L_{\mathrm{ssecl}} + L_{\mathrm{smc}} + L_{\mathrm{il}}, & \text{if } t > T \end{cases} \tag{14}$$

where T marks the start of incremental learning. The overall pseudocode of the IMTCLN is presented in Algorithm 1.

Algorithm 1 IMTCLN

Require: Labeled training set $\mathbf{X}^L$ with labels $\mathbf{Y}$, Unlabeled training set $\mathbf{X}^U$, Number of epochs T.

Build graphs through $\{\mathbf{X}^L, \mathbf{X}^U\}$
Initialize the parameters of the network
for epoch $t = 1$ to T **do**
  Feature Extraction:
  Obtian feature $\mathbf{F}_{\mathrm{I}}$ and $\mathbf{F}_{\mathrm{II}}$ by Eq. (1).
  Obtian the output probabilities of the three classifiers by Eq. (2) and Eq. (10).
  Supervised Classification Task:
  Compute cross-entropy loss by Eq. (3).
  Semi-Supervised Euclidean Contrastive Learning Task:
  Compute contrastive loss by Eq. (4).
  Supervised Mahalanobis Classification Task:
  Compute distances between samples by Eq. (5).
  Compute cross-entropy loss by Eq. (7) .
  if $t \ge T_{\mathrm{start}}$ **then**
    Incremental Learning Task:
    Identify unknown class samples by Eq. (9) and Eq. (11-12).
    Compute cross-entropy loss by Eq. (13).
  end if
  Update model parameters by Eq. (14).
end for
Output OSC results through $\mathbf{P}_{\mathrm{I}}$.
Return: $\mathbf{P}_{\mathrm{I}}$

III. Datasets and Experimental Setup

A. Datasets

In order to fully verify the effectiveness of our method and demonstrate its effectiveness in open-set HSIC with few-shot, we conducted experiments on three common datasets: Indian Pines (IP), Salinas (SA), and University of Pavia (UP). At the same time, the hardware environment of our experiment is equipped with the NVIDIA GeForce A5000 GPU. In addition, Python-3.9.18 language and PyTorch-2.0.1 open-source software framework are used to implement the proposed algorithm program.

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TABLE I
NUMBER OF TRAINING AND TEST DATA ON THREE DATASETS

Indian Pines				Salinas Velley				University of Pavia			
Index	Class Name	Training	Testing	Index	Class Name	Training	Testing	Index	Class Name	Training	Testing
1	Alfalfa	5	41	1	Weeds-1	5	2004	1	Asphalt	5	6626
2	Corn-notill	5	1423	2	Weeds-2	5	3721	2	Meadows	5	18644
3	Corn-mintill	5	825	3	Fallow	5	1971	3	Gravel	5	2094
4	Corn	5	232	4	Fallow-P	5	1389	4	Trees	5	3059
5	Grass-pasture	5	478	5	Fallow-S	5	2673	5	Metal-sheet	5	1340
6	Grass-trees	5	725	6	Stubble	5	3954	6	Bare-soil	5	5024
7	Grass-pasture	5	23	7	Celery	5	3574	7	Bitumen	5	1325
8	Hay-windrowed	5	473	8	Grapes	5	11266	8	Brick	5	3677
9	Oats	5	15	9	Soil	5	6198		Unknown	0	947
10	Soybean-notill	5	967	10	Corn	5	3273				
11	Soybean-mintill	5	2450	11	Lettuce-4wk	5	1063				
12	Soybean-clean	5	588	12	Lettuce-5wk	5	1922				
13	Wheat	5	200	13	Lettuce-6wk	5	911				
14	Woods	5	1260	14	Lettuce-7wk	5	1065				
15	Buildings	5	381		Unknown	0	9075				
	Unknown	0	93								

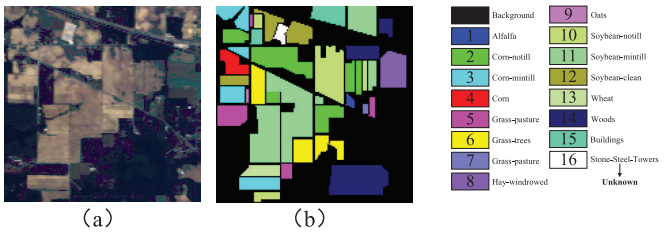


Fig. 5. (a) False-color image and (b) ground truth of the IP dataset.

1) Indian Pines: The IP dataset is a hyperspectral dataset captured by the AVIRIS sensor, consisting of 145 × 145 pixels and 224 spectral bands ranging from 0.4 to 2.5 mm. It contains 16 classes with a total of 10 249 labeled samples, representing an agricultural area with crops of regular geometry and some irregular forest zones. In this article, we set "Stone-Steel-Towers" as the unknown class, with the remaining classes as known classes, and all background pixels do not participate in the experiment. The number of each class is presented in Table I, and the false-color image and ground truth of IP are shown in Fig. 5.

2) SA Valley: The SA scene is a hyperspectral dataset collected by the 224-band AVIRIS sensor over SA Valley, California. It features high spatial resolution (3.7-m pixels) and covers an area of 512 lines by 217 samples, with a total of 54 129 labeled samples. The dataset includes vegetables, bare soils, and vineyard fields, with a total of 16 classes. In this article, we set "Vinyard_untrained" and "Vinyard_Vertical_trellis" as unknown classes, with the remaining classes as known classes, and all background pixels do not participate in the experiment. The number of each class is presented in Table I, and the false-color image and ground truth of SA are shown in Fig. 6.

3) University of Pavia: The Pavia University dataset is a widely used HSI dataset captured over Pavia, Italy, using the ROSIS-3 sensor. It comprises 610 × 340 pixels with 115 spectral bands and is divided into nine classes with a total of 42 776 labeled samples. In this article, we set "Shadows" as the unknown class, with the remaining classes as known classes, and all background pixels do not participate in the experiment. The number of each class is presented in Table I, and the false-color image and ground truth of UP are shown in Fig. 7.

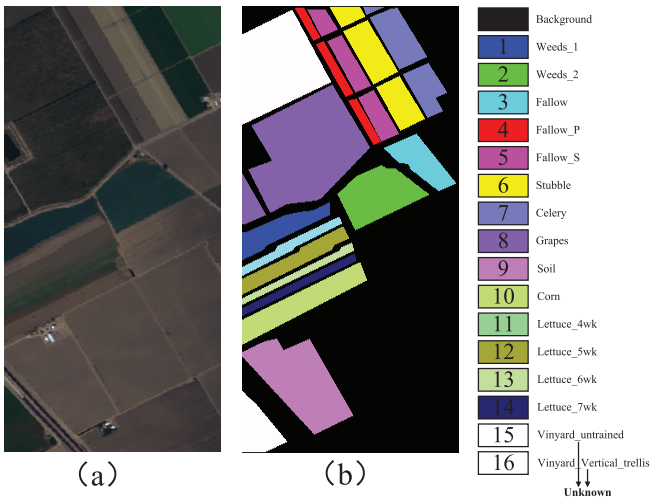


Fig. 6. (a) False-color image and (b) ground truth of the SA dataset.

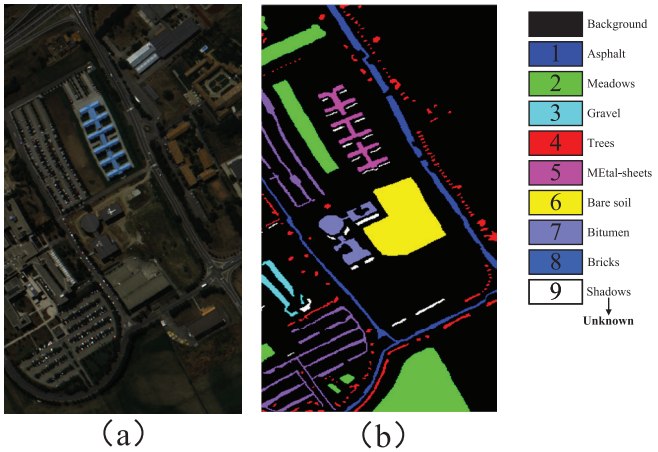


Fig. 7. (a) False-color image and (b) ground truth of the PU dataset.

To validate the performance of our proposed OSC methods of HSI, we select L (L = 5) samples from each class as labeled samples A and the remaining samples from each

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class, along with the unknown class, are used as the test set B, and the background pixels in the dataset are masked. To ensure the reliability of the results, we conducted ten repeated experiments on all datasets and recorded the average accuracy (AA) and standard deviation.

In this article, we set the patch size s to 5, γ to 1.3, and α to 0.01. For the PU and SA datasets, the model was trained for 300 epochs with incremental learning starting at epoch 150, using a batch size of 100. For the IP dataset, the model was trained for 500 epochs with incremental learning beginning at epoch 250, using a batch size of 200. In the actual training process, labeled samples participate in training through feature extractor I and classifier I, while unlabeled samples participate in training through feature extractor II, classifier II, and classifier III. The learning rates were set to 7×10^{-4} for IP, 3×10^{-4} for PU, and 8×10^{-5} for SA. We employed the Adam optimizer with Kaiming initialization and set all loss weights to 1 to maintain training stability and simplify the optimization process.

On this basis, the overall accuracy (OA), the AA, and the Kappa coefficient based on the confusion matrix (100× Kappa) are selected as evaluation indicators.

B. Convergence Analysis

The experiment in Fig. 8 was performed based on training data of five labeled samples in each class. The first row is the cross-entropy loss L_{sc} of the supervised classification task, the second row is the contrastive loss L_{ssecl} of the semi-supervised Euclidean contrastive learning task, the third row is the cross-entropy loss L_{smc} of the supervised Mahalanobis classification task, the fourth row is the cross-entropy loss L_{il} of the incremental learning task, and the last row is the prediction OA of the test set, and the prediction accuracy of the unknown class (Unknown). The x-axis represents the number of epochs. As can be seen from Fig. 8, in the three datasets, not only all four task learning have smooth convergence curves, but also the classification accuracy gradually increases as the loss function decreases. In addition, after starting the incremental learning task, the network can still re-enter the stable phase. This proves that the four proposed learning tasks are not contradictory and can achieve consistent convergence.

It should be noted that in the initial stage of the incremental learning task, the classification accuracy and the loss curves of the other three learning tasks fluctuated dramatically. This is because the unknown class samples detected by the UCR strategy had errors that affected the convergence and accuracy of the network. However, as the network continues to learn under the four learning tasks, not only the detection accuracy of unknown classes is improved, but also OA is further improved, and the four learning tasks tend to be stable again. This proves the effectiveness of our proposed four learning tasks in open-set classification.

C. Experiment for Comparison

1) Experiment Settings: In order to fully evaluate the performance of our proposed IMTCLN algorithm, two traditional algorithms (including SVM and RF), three state-of-the-art few-shot CSC algorithms (including CMFSL [31], FSCF-SSL [30], DMCM [38]), and five latest OSC algorithms (including MDL4OW [44], SSLR [47], SSMLP-RPL [49], LDCLNet [56], EGWA-OSDA [57]) were compared. To ensure the fairness of the experiment, for the CSC algorithm, we detect samples with a probability less than 0.5 threshold as unknown classes. For the traditional classification methods, SVM and RF, we use the classic scikit-learn library for implementation. For the few-shot CSC methods CMFSL, FSCF-SSL, and DMCM, the experimental settings are consistent with the default values in the original open-source code and original file. Since the MDL4OW and SSLR OSC uses the extreme value theory, it is necessary to set the trailing value according to the number of training samples. We use binary search to confirm the optimal trailing value. For the IP, SA, and UP datasets, MDL4OW parameter τ is set to 2, 4, 3, respectively, while the SSLR parameter $\mathcal{T}$ is set to 2, 4, 3, respectively. At the same time, as the number of training samples is set to 5, the epochs of SSMLP-RPL and LDCLNet also need to be adjusted to 200 and 500, and the threshold t of EGWA-OSDA is 0.5 to achieve the best algorithm performance.

2) Experiment Results: To comprehensively validate the effectiveness of the proposed method, this study conducts comparative experiments on three representative hyperspectral datasets: IP, SA, and PU. Furthermore, to verify the robustness of the method under different open-set scenarios, the experiments are configured with unknown class sample ratios of 0.9%, 16.7%, and 2.2% across the three datasets, representing varying degrees of openness. It is proved that IMCLT has good recognition ability for both a small number and a large number of unknown classes, and can also accurately identify unknown classes in typical open environments. Tables II–IV present the quantitative evaluation results for each dataset, while Figs. 9–11 display the corresponding classification visualization results.

It can be seen from Tables II–IV that on the IP, SA, and PU datasets, respectively, the IMTCLN method is at least 6.61%, 10.47%, and 7.89% higher than other methods in OA; at least 11.33%, 5.02%, and 14.99% higher in AA; at least 7.02%, 11.83%, and 10.84% higher in Kappa coefficient; and at least 0.75%, 70.43%, and 30.62% higher in unknown class accuracy. The experimental results demonstrate that the IMTCLN method exhibits significant advantages in few-shot open-set scenarios.

For other comparison algorithms, traditional classification methods (SVM, RF) showed unsatisfactory classification results on the three datasets due to their reliance on linear transformation and lack of an open-set recognition mechanism. In addition, although the few-shot closed-set methods (CMFSL, DMCM) enhance the transferability of features through meta-learning, the recognition accuracy of unknown classes is still less than 1% due to the closed-set assumption, which limits their practicality. Otherwise, FSCF-SSL achieves 40.57% unknown class accuracy on the PU dataset, as the method is able to learn class-invariant features from the data itself. However, its performance is unstable, with unknown class accuracy below 6% on the IP and SA datasets. Existing open-set methods (MDL4OW, SSLR, SSMLP-RPL,

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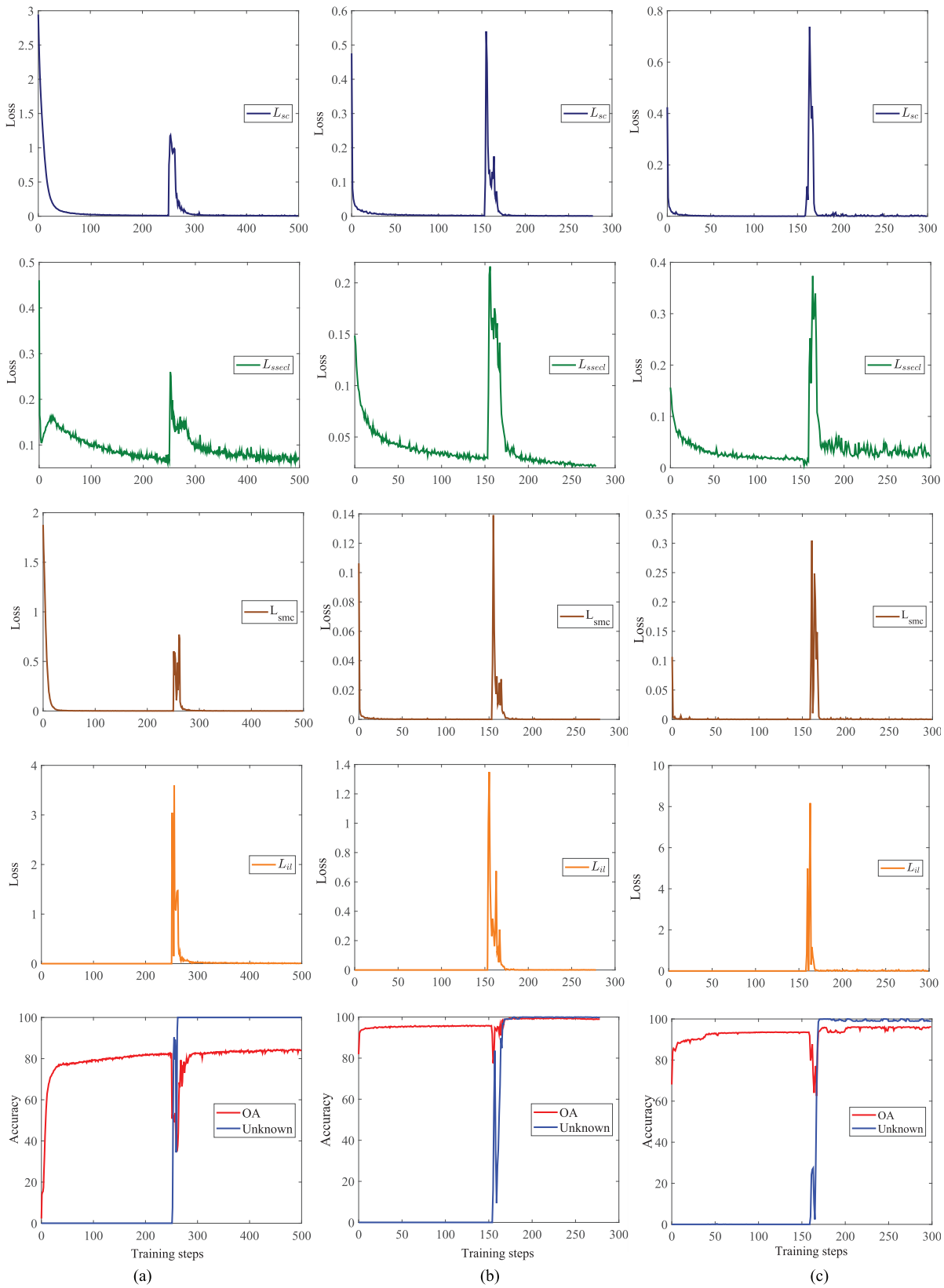


Fig. 8. Loss curves of the IMTCLN and the predicted accuracy of the testing set on three datasets (five labeled samples per class). (a) IP. (b) SA valley. (c) UP.

LDCLNet, EGWA-OSDA) show unsatisfactory OSC performance under few-shot conditions.

The visualization results in Figs. 9–11 further validate the effectiveness of our method. Traditional methods fail to

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TABLE II
THEMATIC CLASSIFICATION MAPS OF DIFFERENT METHODS ON THE IP SCENE

Class	SVM	RF	CMFSL	FSCF-SSL	DMCM	MDL4OW	SSLR	SSMLP-RPL	LDCLNet	EGWA-OSDA	OURS
1	79.57±8.43	80.65±9.25	98.54±1.95	**100.00±0.00**	**100.00±0.00**	67.61±12.00	91.96±15.53	84.78±10.20	89.13±8.70	97.39±4.22	99.69±0.76
2	43.64±10.95	38.99±11.81	45.65±7.34	47.63±13.70	60.98±7.68	39.31±12.29	47.45±9.93	37.42±8.48	38.62±15.86	43.89±7.75	**69.70±12.52**
3	37.69±11.17	42.53±7.37	63.30±7.57	63.39±15.08	**68.56±8.89**	44.12±8.76	48.08±11.32	39.66±8.43	35.60±1.63	52.67±11.19	69.66±16.18
4	51.65±7.37	54.43±13.23	85.00±8.22	90.34±9.25	90.26±2.73	72.62±12.59	58.44±17.62	66.41±14.48	70.25±1.05	89.20±7.73	**99.82±0.44**
5	60.10±11.64	63.02±6.15	80.46±7.99	70.46±14.59	79.75±2.05	69.42±11.02	60.37±15.44	57.25±9.01	67.45±4.23	86.66±2.03	**83.08±8.00**
6	80.73±8.98	78.41±11.06	86.68±6.49	64.41±17.93	94.84±1.67	73.42±19.19	89.15±7.63	79.42±9.74	77.06±7.88	89.51±2.68	**99.94±0.10**
7	95.36±4.24	93.57±3.11	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	85.71±9.97	60.00±48.99	82.14±7.82	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
8	72.32±9.46	62.82±9.06	88.63±7.39	95.35±9.20	98.56±0.45	78.81±12.41	90.21±8.20	80.00±13.61	83.368±6.38	88.87±10.45	**100.00±0.00**
9	97.00±5.10	95.50±6.10	**100.00±0.00**	**100.00±0.00**	97.33±3.27	**100.00±0.00**	39.50±48.40	96.00±5.83	**100.00±0.00**	99.00±2.00	**100.00±0.00**
10	43.88±9.68	49.92±4.73	66.08±3.30	54.27±10.75	53.42±3.71	58.45±11.29	64.00±11.96	65.78±14.01	51.96±4.73	62.49±11.58	**86.48±5.79**
11	31.07±9.38	38.13±6.70	57.67±17.79	61.66±19.37	**77.13±1.51**	52.16±10.29	51.14±11.03	48.11±4.53	42.04±0.94	50.62±17.07	72.17±15.54
12	37.30±9.42	32.92±4.55	47.86±12.47	63.71±18.01	84.22±2.56	43.39±12.54	47.08±10.94	43.00±5.80	53.04±17.12	61.05±17.63	**86.70±10.59**
13	93.32±3.60	92.49±6.05	99.00±1.30	99.50±0.55	99.70±0.24	82.83±17.28	96.00±6.01	81.56±10.43	96.83±2.68	98.63±1.36	**99.93±0.17**
14	60.94±14.61	59.12±9.86	81.76±7.90	87.48±8.69	**94.46±1.32**	79.81±14.08	84.06±8.08	66.12±6.19	72.85±12.77	83.51±6.99	93.21±9.80
15	54.02±9.57	40.08±5.34	75.91±14.85	87.93±11.48	76.43±3.13	60.41±15.03	63.47±7.46	66.94±7.72	52.20±2.72	87.41±5.52	**96.71±8.07**
Unknown	0.00±0.00	0.00±0.00	0.43±0.53	2.15±3.79	0.00±0.00	0.22±0.43	99.25±1.36	54.19±43.16	1.08±1.08	0.00±0.00	**100.00±0.00**
OA	48.27±2.45	48.85±2.41	66.98±4.35	66.76±4.48	76.72±1.10	57.59±3.90	62.54±2.94	56.38±2.62	53.80±4.68	65.11±3.43	**82.83±3.76**
AA	58.66±1.97	57.66±1.76	73.56±1.61	74.27±2.01	79.47±0.50	63.00±2.72	68.14±8.32	66.19±3.23	63.83±1.78	74.43±1.12	**91.07±1.25**
Kappa	42.82±2.56	43.22±2.42	61.97±4.60	62.57±4.61	73.60±1.23	53.57±4.05	58.20±3.25	49.28±4.12	53.80±4.68	60.96±3.66	**80.62±4.21**

TABLE III
THEMATIC CLASSIFICATION MAPS OF DIFFERENT METHODS ON THE SA SCENE

Class	SVM	RF	CMFSL	FSCF-SSL	DMCM	MDL4OW	SSLR	SSMLP-RPL	LDCLNet	EGWA-OSDA	OURS
1	97.82±2.30	98.32±1.23	97.58±1.69	**100.00±0.00**	98.55±1.40	90.30±9.34	80.57±1.02	93.26±6.08	85.49±14.06	99.40±0.50	98.13±2.67
2	97.99±3.54	97.29±3.06	99.70±0.39	98.26±1.44	99.49±0.19	86.34±10.33	85.04±2.68	92.58±6.87	50.90±18.77	**99.99±0.03**	99.95±0.09
3	96.86±2.02	83.68±14.05	96.46±3.55	96.84±4.44	**100.00±0.00**	66.88±15.78	90.73±11.54	78.87±15.18	71.43±0.58	97.29±2.87	98.11±5.61
4	98.78±0.70	96.43±2.46	97.38±1.57	**99.88±0.17**	99.50±0.19	80.78±17.31	75.31±6.63	90.31±7.03	91.32±7.82	99.54±0.30	98.24±3.96
5	93.61±3.56	95.77±3.70	95.41±0.98	97.27±1.52	93.49±2.18	91.78±10.77	71.61±3.51	89.85±3.44	52.56±0.80	96.97±0.95	**98.16±1.68**
6	96.67±2.44	98.71±0.59	99.02±1.52	99.84±0.11	99.95±0.98	88.25±10.09	68.95±4.42	95.62±1.17	99.91±0.06	99.63±0.61	**100.00±0.00**
7	99.00±0.26	98.12±2.55	99.17±0.97	99.69±0.21	98.77±0.99	84.43±16.48	75.14±1.67	94.42±4.37	97.30±0.82	99.64±0.27	**99.99±0.01**
8	95.90±1.72	96.44±1.46	93.88±2.81	**97.86±0.77**	95.79±2.60	79.65±18.36	84.57±1.34	86.01±11.23	63.29±20.82	88.64±7.44	78.91±16.05
9	99.39±0.33	97.06±3.02	99.48±0.60	99.08±0.91	99.98±0.03	79.77±17.32	95.25±1.47	89.66±6.37	93.23±6.30	98.14±1.46	**100.00±0.00**
10	82.95±9.37	80.72±8.08	86.34±3.69	91.20±7.28	77.63±1.49	69.54±12.33	57.96±7.26	79.71±14.67	62.11±2.41	88.64±3.34	**93.26±6.46**
11	91.69±2.36	93.01±3.49	97.06±1.65	96.26±4.19	99.21±0.37	90.02±8.54	82.96±1.28	87.81±9.93	71.91±5.15	93.89±5.30	**99.46±0.28**
12	99.37±0.56	90.78±7.91	98.24±2.93	98.21±1.45	99.95±0.08	89.91±6.53	99.01±1.65	96.50±2.50	96.78±0.52	**100.00±0.00**	**100.00±0.00**
13	96.03±6.02	92.42±5.83	99.93±0.05	99.78±0.16	99.93±0.09	82.22±12.51	81.58±7.30	87.62±8.37	89.79±1.04	**100.00±0.00**	99.72±0.12
14	92.78±4.08	82.94±9.08	**99.47±0.53**	98.14±1.20	97.33±0.92	79.22±15.92	76.01±10.26	87.75±14.79	93.65±0.84	95.59±5.04	97.95±0.73
Unknown	15.41±5.90	2.67±1.86	0.22±0.14	5.75±4.97	0.23±0.00	24.18±21.92	17.30±5.34	9.67±9.04	13.27±8.75	29.34±13.18	**99.77±0.66**
OA	82.47±0.80	79.14±0.52	80.30±0.50	82.51±0.76	81.00±0.41	72.32±2.64	70.63±1.27	75.74±1.12	66.07±2.60	84.36±0.73	**94.83±3.39**
AA	84.64±0.68	81.52±1.01	90.62±0.33	91.87±0.36	91.35±0.24	78.88±3.08	76.13±2.69	89.98±1.88	75.53±0.70	92.42±0.88	**97.44±1.34**
kappa	80.18±0.91	76.48±0.55	77.81±0.51	80.20±0.86	78.59±0.48	68.89±3.13	66.37±1.51	72.45±1.29	62.54±2.22	82.37±0.82	**94.20±3.80**

TABLE IV
THEMATIC CLASSIFICATION MAPS OF DIFFERENT METHODS ON THE UP SCENE

Class	SVM	RF	CMFSL	FSCF-SSL	DMCM	MDL4OW	SSLR	SSMLP-RPL	LDCLNet	EGWA-OSDA	OURS
1	53.96±6.45	45.85±11.65	63.48±28.67	73.29±16.06	**83.61±5.08**	60.14±17.51	74.12±12.37	59.08±23.33	63.69±4.27	80.61±6.93	80.98±16.21
2	62.48±10.74	65.97±12.24	72.05±15.19	81.00±8.03	**96.88±1.30**	57.17±12.46	76.08±4.55	51.28±14.59	63.67±13.41	60.33±9.88	96.96±3.90
3	58.76±14.74	57.10±12.26	45.44±23.07	63.14±23.48	55.18±3.12	57.52±12.46	71.08±9.96	63.84±17.89	25.56±24.39	76.55±7.62	**99.81±0.33**
4	80.99±13.02	84.57±5.58	63.65±28.47	**93.25±4.64**	86.17±4.29	78.45±10.97	88.31±7.30	70.35±26.35	67.59±7.31	93.85±4.54	76.43±13.13
5	86.32±21.79	97.80±1.70	73.10±35.07	99.82±0.25	99.55±0.57	99.25±1.06	39.55±13.06	91.35±12.31	99.22±0.56	97.28±5.19	**100.00±0.00**
6	55.79±15.12	55.40±13.79	57.95±24.45	85.62±4.08	63.93±5.06	66.14±5.77	66.27±18.69	62.08±18.57	31.77±10.03	81.76±11.25	**99.68±0.55**
7	67.42±14.38	63.92±11.76	53.45±30.32	75.05±15.02	95.83±1.91	80.25±8.17	84.35±8.51	74.80±21.28	37.18±28.61	98.51±0.82	**100.00±0.00**
8	52.66±13.98	55.55±10.56	71.84±5.82	89.54±8.04	80.31±4.78	57.39±15.81	72.38±8.77	61.12±12.10	42.29±30.15	79.59±7.66	**84.52±22.26**
Unknown	0.00±0.00	0.00±0.00	3.18±3.56	40.57±42.39	4.54±3.25	59.44±28.07	39.18±30.56	67.24±14.11	28.14±18.32	0.00±0.00	**97.86±2.09**
OA	60.19±2.68	61.08±5.90	65.83±16.11	80.60±5.05	84.72±1.39	62.34±7.81	73.23±2.83	58.93±5.58	56.00±10.16	71.86±3.21	**92.61±1.63**
AA	57.60±2.73	58.46±3.39	56.02±16.95	77.92±1.79	74.00±0.87	68.42±4.97	67.92±2.64	66.79±3.90	51.01±8.59	74.28±1.71	**92.91±0.92**
Kappa	50.07±2.09	51.23±6.37	55.74±18.92	75.47±6.00	79.46±1.83	54.57±8.35	65.67±3.82	51.14±5.46	46.89±11.37	65.26±3.42	**90.30±2.10**

achieve satisfactory classification outcomes on any dataset. Few-shot closed-set approaches misclassify unknown class samples as known classes, while current open-set methods exhibit inconsistent accuracy in identifying unknown class

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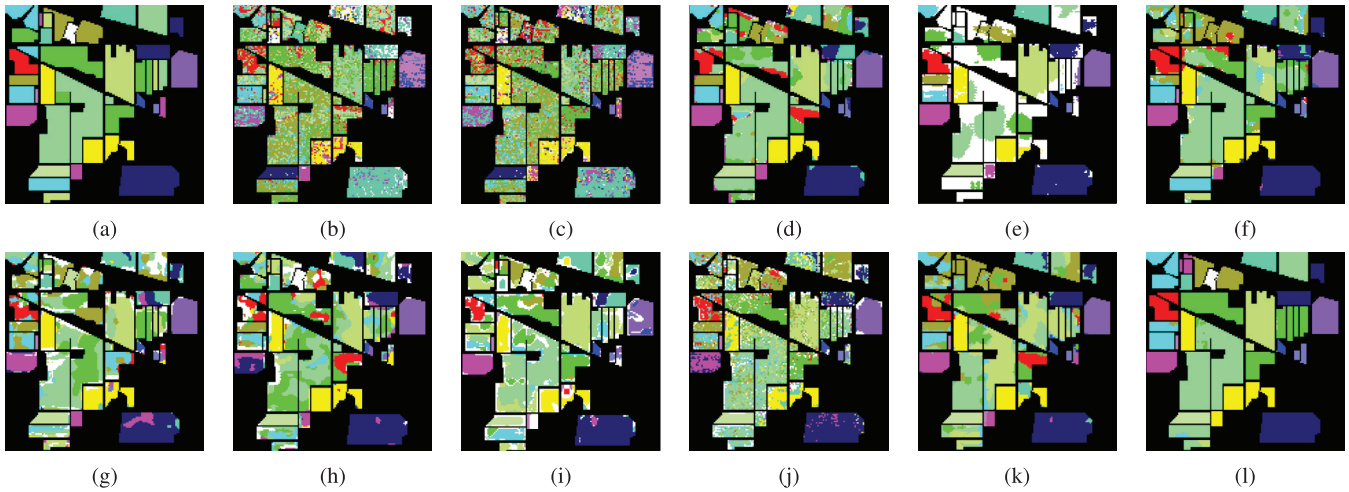


Fig. 9. Classification maps obtained by different methods on the IP dataset with five training samples. (a) GT. (b) SVM. (c) RF. (d) CMFSL. (e) FSCF-SSL. (f) DMCM. (g) MDL4OW. (h) SSLR. (i) SSMLP-RPL. (j) LDCLNet. (k) EGWA-OSDA. (l) OURS.

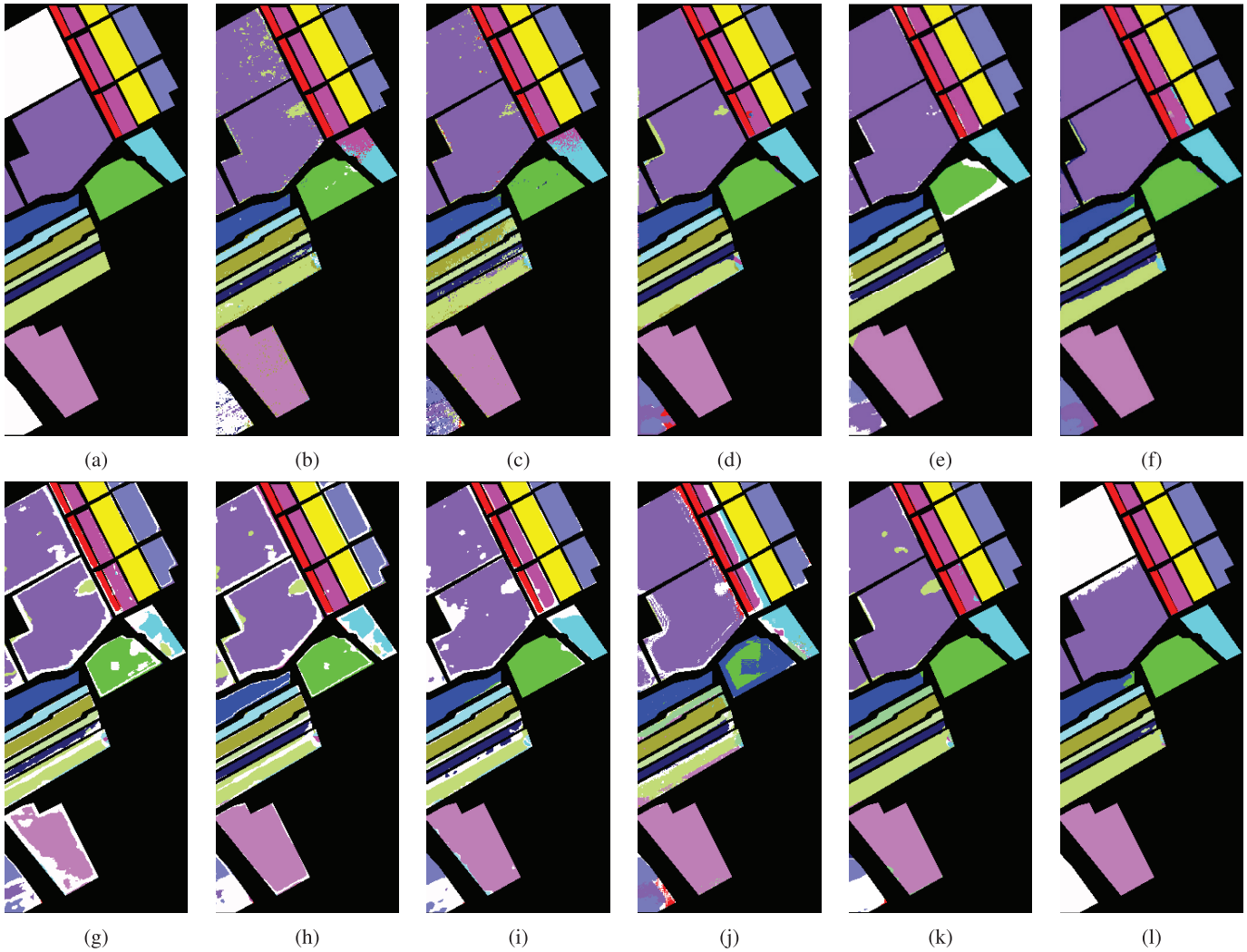


Fig. 10. Classification maps obtained by different methods on the SA dataset with five training samples. (a) GT. (b) SVM. (c) RF. (d) CMFSL. (e) FSCF-SSL. (f) DMCM. (g) MDL4OW. (h) SSLR. (i) SSMLP-RPL. (j) LDCLNet. (k) EGWA-OSDA. (l) OURS.

samples. In contrast, IMTCLN generates classification maps with optimal spatial consistency to reference images through multitask constraints, particularly maintaining sharper distinctions at class boundary regions.

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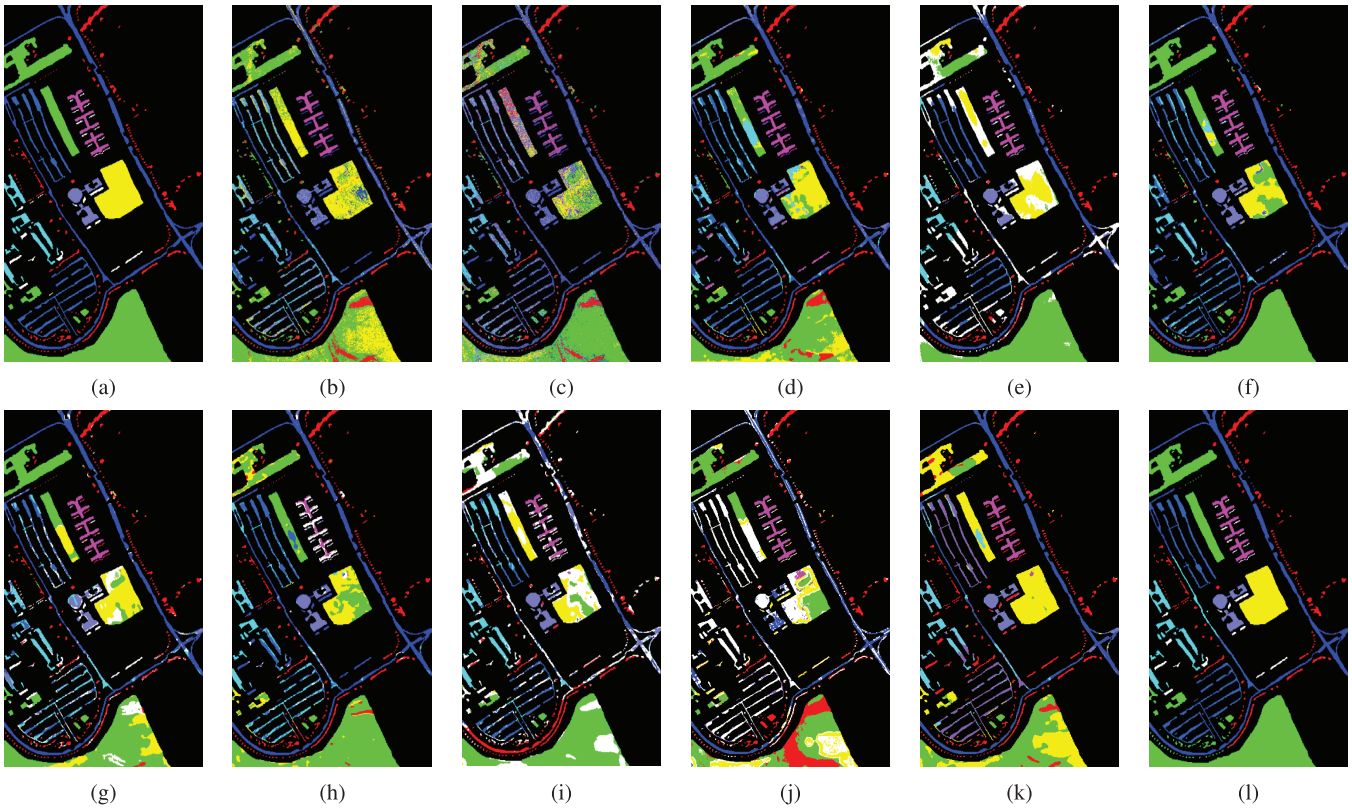


Fig. 11. Classification maps obtained by different methods on the PU dataset with five training samples. (a) GT. (b) SVM. (c) RF. (d) CMFSL. (e) FSCF-SSL. (f) DMCM. (g) MDL4OW. (h) SSLR. (i) SSMLP-RPL. (j) LDCLNet. (k) EGWA-OSDA. (l) OURS.

TABLE V
ABLATION EXPERIMENTS OF THE PROPOSED ALGORITHM ON THREE DATASETS

		(a)	(b)	(c)	(d)	(e)	(f)	(g)
L_{sc}		√	√	√	√	√	√	√
L_{ssecl}		×	√	√	×	√	√	√
L_{smc}		×	×	×	√	√* (ED)	√	√
L_{il}		×	×	√	√	√	× *	√
IP	OA	54.72±4.38	77.22±4.82	77.19±7.74	56.97±5.08	79.45±3.58	76.45±4.37	**82.83±3.76**
	AA	62.24±2.36	82.99±2.54	87.24±3.67	69.29±2.79	87.33±4.28	80.32±2.31	**91.07±1.25**
	F1	55.26±4.32	77.12±6.12	77.33±7.49	58.25±5.05	80.26±4.36	75.98±6.30	**83.21±3.92**
	Kappa	49.20±4.78	74.48±5.34	74.45±7.36	51.87±5.27	76.83±4.15	73.35±4.84	**80.62±4.21**
	Un_acc	0.00±0.00	0.00±0.00	99.78±0.43	98.28±1.46	79.57±39.79	0.00±0.00	**100.00±0.00**
SA	OA	76.97±0.69	82.77±0.39	82.24±6.39	79.99±2.54	94.40±4.62	82.25±0.70	**94.83±3.39**
	AA	85.27±0.89	92.51±0.7	91.20±4.88	89.95±1.14	97.03±2.07	92.09±0.85	**97.44±1.34**
	F1	70.85±0.79	76.63±0.44	79.90±8.06	78.7±3.04	94.44±4.56	76.05±0.48	**94.73±3.54**
	Kappa	74.11±0.72	80.56±0.43	80.02±7.33	77.59±2.85	93.72±5.17	79.96±0.83	**94.20±3.80**
	Un_acc	0.00±0.00	0.00±0.00	90.51±23.67	92.87±3.12	95.85±7.95	0.00±0.00	**99.95±0.09**
UP	OA	69.74±6.49	91.38±4.06	85.50±4.02	70.40±3.85	82.01±6.57	87.44±6.90	**92.61±1.63**
	AA	69.5±2.63	85.32±1.62	85.56±2.01	71.75±0.61	82.33±6.87	81.12±3.83	**92.91±0.92**
	F1	69.52±5.95	91.5±2.61	89.42±14.17	71.42±2.35	77.29±26.66	87.13±6.40	**89.26±13.47**
	Kappa	62.26±6.67	88.82±4.9	81.25±4.69	62.60±3.30	77.23±7.60	83.73±8.54	**90.30±2.10**
	Un_acc	0.00±0.00	0.00±0.00	76.21±25.26	83.16±24.8	77.98±25.47	4.69±9.32	**97.86±2.09**

D. Ablation Experiment

In order to demonstrate the necessity and effectiveness of different learning tasks in the proposed IMTCLN method, we conducted ablation experiments on the loss functions corresponding to the four learning tasks on the IP, SA, and UP datasets. The ablation experiments are set as Table V, we conduct seven ablation studies: (a) only supervised classification loss L_{sc}, (b) supervised classification loss L_{sc} combined with semi-supervised Euclidean contrastive loss L_{ssecl}, (c) without supervised Mahalanobis classification loss L_{smc}, (d) without semi-supervised Euclidean contrastive loss L_{ssecl}, (e) the Mahalanobis distance is changed to Euclidean distance,

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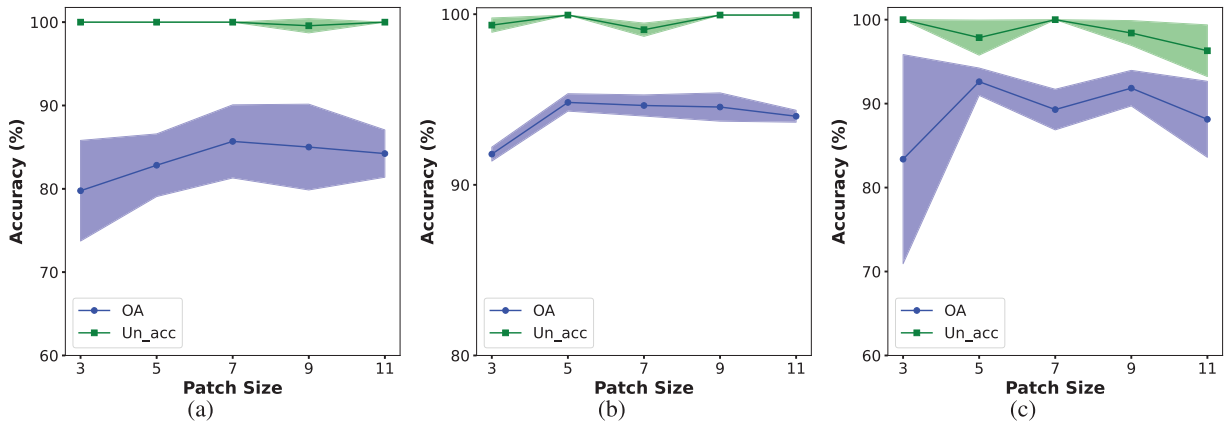


Fig. 12. Impact of patch size on classification accuracy. (a) IP dataset. (b) SA dataset. (c) PU dataset.

(f) the incremental learning is replaced by judging samples with prediction values less than 0.5 as unknown classes, and (g) the full model. In order to eliminate randomness in the experiments, each experiment is repeated ten times, and the mean and standard deviation are reported.

In the ablation study, the comparison between experiments (a) and (b) shows that the introduction of the L_{ssecl} loss can improve OA by 22.5%, 5.8%, and 21.64% across three datasets, demonstrating that the L_{ssecl} loss effectively enhances classification accuracy for known classes. In addition, from the ablation experiment (d), we can see that when IMTCLN does not learn the semi-supervised Euclidean contrastive learning task, its unknown class accuracy decreases by 1.72%–14.7%. This proves that the semi-supervised Euclidean contrastive learning task can further improve the recognition performance of unknown classes through the collaborative incremental learning task. The reason is that the incremental learning task introduces the unknown pseudo-label learning in the test set, and the semi-supervised Euclidean comparative learning task can realize the label propagation in the test set, which makes the correct unknown pseudo-labels propagate quickly in the network, and the wrong unknown pseudo-labels are corrected in the network.

Comparative analysis of experiments (c) and (g) shows that when IMTCLN does not learn the supervised Mahalanobis classification task, the unknown class accuracy decreases by 0.22%–21.65%, while the standard deviation increases by 0.43%–23.58%. This indicates that omitting the supervised Mahalanobis classification task training will have an adverse impact on the accuracy and stability of the UCR.

From the ablation experiment (e), we can see that when IMTCLN uses Euclidean distance to measure unknown classes, the accuracy of unknown classes decreases by 4%–20%, and the standard deviation increases by 7%–39%. This verifies the advantage of the Mahalanobis metric over Euclidean distance in decoupling feature correlation and improving the recognition accuracy of unknown samples.

The data in the ablation experiment (f) show that when IMTCLN does not add the incremental learning task, the detection accuracy of unknown classes is less than 5%. This shows that the network's detection ability for unknown classes is insufficient at this time, proving the necessary role of incremental learning in realizing the unknown class recognition function.

In summary, through the experimental data of ablation experiments and data analysis, the following can be proved.

1) Semi-supervised Euclidean contrastive learning task can effectively improve the accuracy of OSC.
2) The proposed supervised Mahalanobis classification task can reduce the correlation between the feature vectors to improve the accuracy of the UCR.
3) The incremental learning strategy can effectively improve the recognition accuracy and stability of unknown classes. At the same time, it can be combined with semi-supervised Euclidean contrastive learning tasks to correct label propagation, thereby achieving better results.

E. Analysis of Patch Size

The size of the patch significantly influences the feature extraction and information propagation of the graph in HSIs. Fig. 12 shows the effect of different patch sizes on classification accuracy. The results reveal that the classification accuracy for known classes initially increases and then decreases with larger patch sizes, while the accuracy for unknown classes remains relatively stable. The reason for this is that a larger patch size can provide more spatial context information, as the patch size continues to increase, noise will be introduced, and performance will degrade. To balance classification accuracy and computational efficiency, we ultimately set s to 5 as the optimal configuration for the network in the three datasets.

F. Analysis of Computational Complexity

In this article, we conduct an evaluation of computational costs across benchmark methods, measuring parameter count (M), training time (s), and inference time (s). As summarized in Table VI, our method achieves the lowest parameter count and fastest inference speed among all methods, albeit with extended training duration. This overhead is primarily

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TABLE VI
COMPUTATIONAL COMPLEXITY OF DIFFERENT METHODS

		CMFSL	FSCF-SSL	DMCM	MDL4OW	SSLR	SSMLP-RPL	LDCLNet	EGWA-OSDA	OURS
IP	Params(M)	0.65	3.31	0.24	0.42	0.37	10.46	14.74	1.91	0.21
	Training time(s)	1997.29	1291.42	3352.19	70.19	134.61	2569.61	1894.63	326.09	2396.53
	Testing time(s)	1.05	5.87	0.83	18.64	14.59	37.49	2.35	1.86	0.15
SA	Params(M)	0.65	3.31	0.24	0.42	0.37	10.46	14.78	1.91	0.21
	Training time(s)	1970.45	2756.86	3388.62	116.54	135.89	1126.71	3133.51	1659.34	5058.83
	Testing time(s)	5.41	25.91	5.35	62.86	62.79	362.62	20.43	14.78	0.22
PU	Params(M)	0.64	3.16	0.04	0.31	0.28	2.93	14.48	1.91	0.12
	Training time(s)	810.49	1106.19	1418.36	75.73	89.27	1086.69	1318.48	595.63	4121.68
	Testing time(s)	3.49	11.97	2.81	81.42	33.86	450.93	11.53	5.36	0.18

attributable to the inclusion of the Mahalanobis distance measurement method, which requires computing a large amount of covariance. It is worth noting that our framework takes the least inference time. The reason is that it can directly learn a unified feature representation of known class boundaries and unknown class rejection capabilities, which realizes end-to-end OSC and has stronger OSC robustness.

IV. CONCLUSION

In this article, a few-shot OSC method for HSIs based on incremental multitask contrastive learning is proposed to solve the dual challenges of scarce labeled samples and unknown class samples. Conventional closed-set methods inherently misclassify unknown class samples as known classes in open-set scenarios, while existing open-set approaches suffer from performance degradation under limited labeled data.

IMTCLN combines supervised classification, semi-supervised Euclidean contrastive learning, supervised Mahalanobis classification, and incremental learning through the feature sharing framework. This multitask coordinated training method significantly improves the adaptability and robustness of the network in data-scarce and open scenarios. Specifically, the semi-supervised Euclidean contrastive learning significantly improves the model's ability to discriminate known classes by mining the potential relationships of unlabeled samples. Second, the supervised Mahalanobis classification task effectively decouples feature correlations and greatly enhances the stability of unknown class recognition. Finally, the incremental learning and semi-supervised Euclidean contrastive learning are collaboratively optimized to achieve more stable unknown class feature learning. This end-to-end network architecture enables the model to directly process raw inputs with high processing efficiency, greatly improving the convenience and efficiency of actual deployment. Experimental verification on three benchmark datasets confirms that IMTCLN consistently outperforms state-of-the-art methods in terms of accuracy and stability for both known and unknown classes.

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